ADVANCE COMPUTER NETWORKS

ASSIGNMENT 7

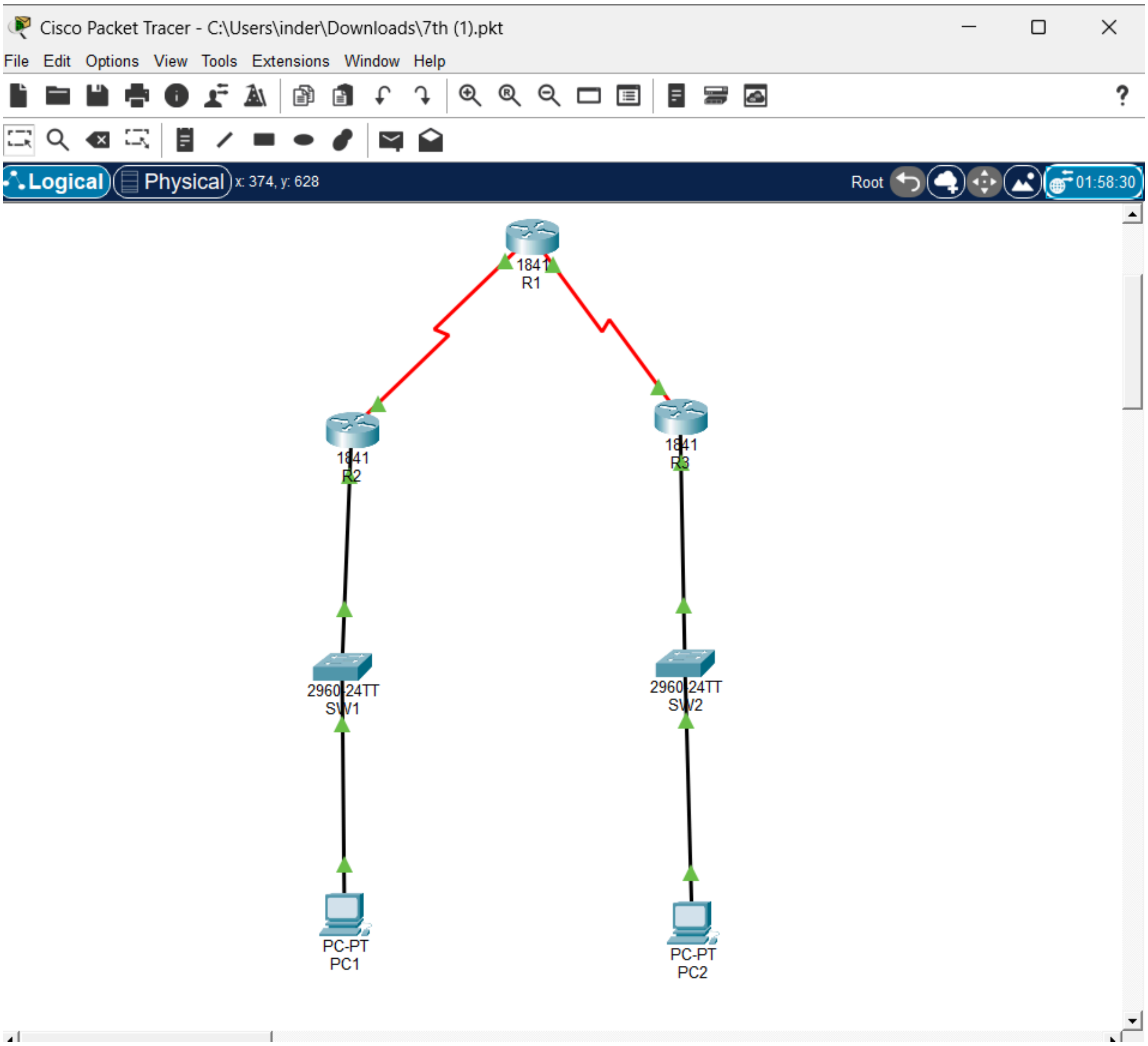
Routing using OSPF (Link State Routing) in Cisco Packet Tracer

Objective:

To understand the Link State Routing algorithm (OSPF)

To configure and verify OSPF routing in a network

To analyze how Shortest Path First (SPF) algorithm works



R1

```
R1>enable
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        unassigned      YES unset  administratively down down
FastEthernet0/1        unassigned      YES unset  administratively down down
Serial0/0/0            100.1.1.1       YES manual up                    up
Serial0/0/1            20.1.1.1        YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
R1#
R1#
```

R2

```
R2>enable
R2#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        172.16.2.1      YES manual up                    up
FastEthernet0/1        unassigned      YES unset  administratively down down
Serial0/0/0            100.1.1.2       YES manual up                    up
Serial0/0/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R2#
R2#
R2#
```

R3

```
R3>enable
R3#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        10.2.2.1        YES manual up                    up
FastEthernet0/1        unassigned      YES unset  administratively down down
Serial0/0/0            20.1.1.2        YES manual up                    up
Serial0/0/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R3#
```

R1

router ospf 1

router-id 1.1.1.1

network 100.1.1.0 0.0.0.255 area 0

network 20.1.1.0 0.0.0.255 area 0

R2

router ospf 1

router-id 2.2.2.2

network 100.1.1.0 0.0.0.255 area 0

network 172.16.0.0 0.0.255.255 area 0

R3

router ospf 1

router-id 3.3.3.3

network 20.1.1.0 0.0.0.255 area 0

network 10.2.2.0 0.0.0.255 area 0

R1

```
R1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/24 is subnetted, 1 subnets
O       10.2.2.0 [110/65] via 20.1.1.2, 00:10:47, Serial0/0/1
     20.0.0.0/24 is subnetted, 1 subnets
C       20.1.1.0 is directly connected, Serial0/0/1
     100.0.0.0/24 is subnetted, 1 subnets
C       100.1.1.0 is directly connected, Serial0/0/0
     172.16.0.0/24 is subnetted, 1 subnets
O       172.16.2.0 [110/65] via 100.1.1.2, 00:10:47, Serial0/0/0
```

R2

```
R2#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/24 is subnetted, 1 subnets
O       10.2.2.0 [110/129] via 100.1.1.1, 00:10:08, Serial0/0/0
     20.0.0.0/24 is subnetted, 1 subnets
O       20.1.1.0 [110/128] via 100.1.1.1, 00:10:08, Serial0/0/0
     100.0.0.0/24 is subnetted, 1 subnets
C       100.1.1.0 is directly connected, Serial0/0/0
     172.16.0.0/24 is subnetted, 1 subnets
C       172.16.2.0 is directly connected, FastEthernet0/0
```

R3

```
R3#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/24 is subnetted, 1 subnets
C       10.2.2.0 is directly connected, FastEthernet0/0
     20.0.0.0/24 is subnetted, 1 subnets
C       20.1.1.0 is directly connected, Serial0/0/0
     100.0.0.0/24 is subnetted, 1 subnets
O       100.1.1.0 [110/128] via 20.1.1.1, 00:09:27, Serial0/0/0
     172.16.0.0/24 is subnetted, 1 subnets
O       172.16.2.0 [110/129] via 20.1.1.1, 00:09:27, Serial0/0/0
```

R1

```
R1#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
2.2.2.2           0   FULL/  -        00:00:38    100.1.1.2       Serial0/0/0
3.3.3.3           0   FULL/  -        00:00:37    20.1.1.2        Serial0/0/1
R1#
```

R2

```
R2#
R2#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           0   FULL/  -        00:00:38    100.1.1.1       Serial0/0/0
R2#
R2#
```

R3

```
R3#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           0   FULL/  -        00:00:39    20.1.1.1        Serial0/0/0
R3#
R3#
```

R1

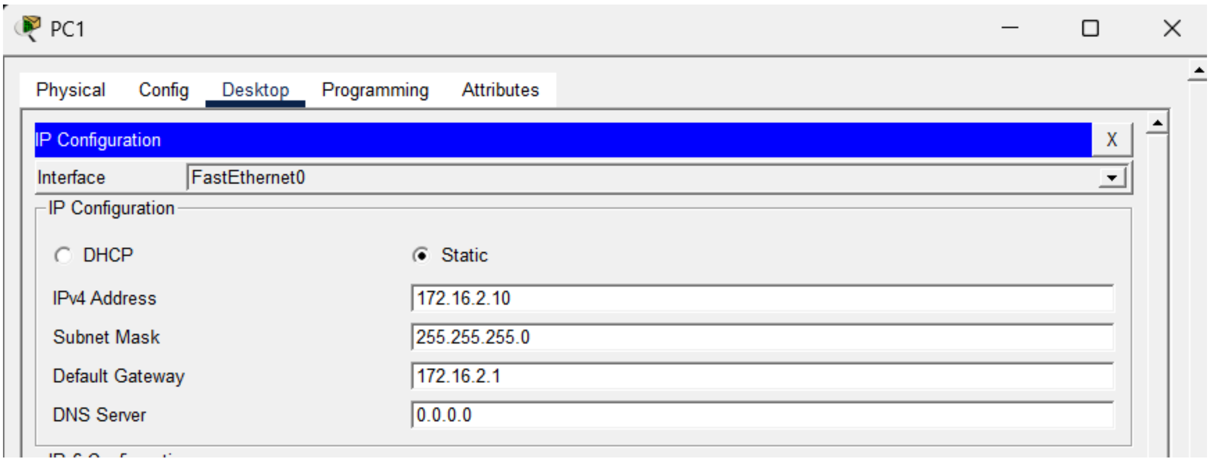
```
R1#show ip ospf database
            OSPF Router with ID (1.1.1.1) (Process ID 1)

                Router Link States (Area 0)

Link ID         ADV Router      Age         Seq#       Checksum Link count
3.3.3.3         3.3.3.3         888         0x80000004 0x008ccd 3
2.2.2.2         2.2.2.2         888         0x80000004 0x00ac64 3
1.1.1.1         1.1.1.1         888         0x80000005 0x007c90 4
R1#
R1#
R1#
```

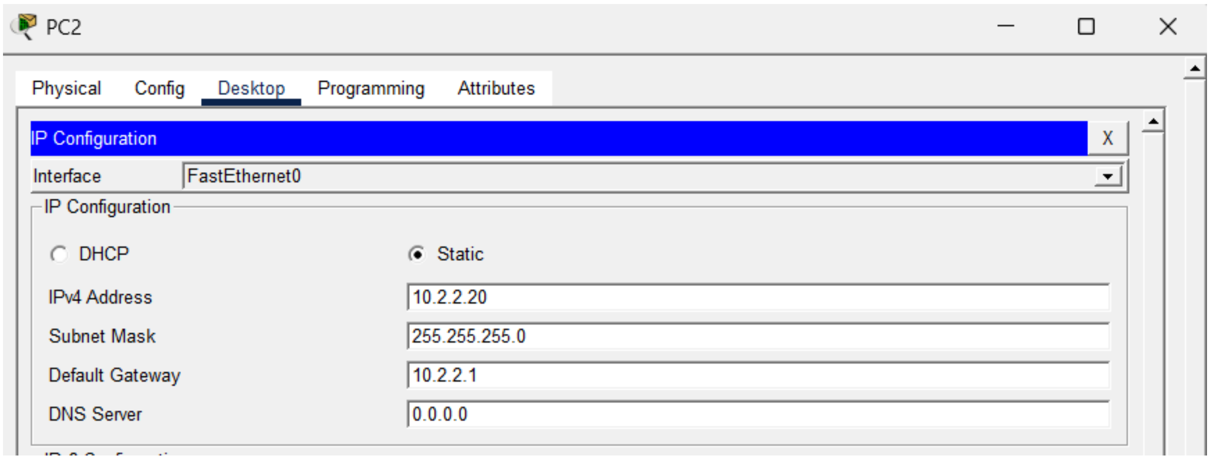
R2

```
R2#show ip ospf database
            OSPF Router with ID (2.2.2.2) (Process ID 1)

                Router Link States (Area 0)

Link ID         ADV Router      Age         Seq#       Checksum Link count
2.2.2.2         2.2.2.2         940         0x80000004 0x00ac64 3
1.1.1.1         1.1.1.1         940         0x80000005 0x007c90 4
3.3.3.3         3.3.3.3         941         0x80000004 0x008ccd 3
R2#
```

R3

```
R3#show ip ospf database
            OSPF Router with ID (3.3.3.3) (Process ID 1)

                Router Link States (Area 0)

Link ID         ADV Router      Age         Seq#       Checksum Link count
3.3.3.3         3.3.3.3         980         0x80000004 0x008ccd 3
2.2.2.2         2.2.2.2         980         0x80000004 0x00ac64 3
1.1.1.1         1.1.1.1         980         0x80000005 0x007c90 4
R3#
```





Pinging PC1 (172.16.2.10) from PC2 (10.2.2.20)